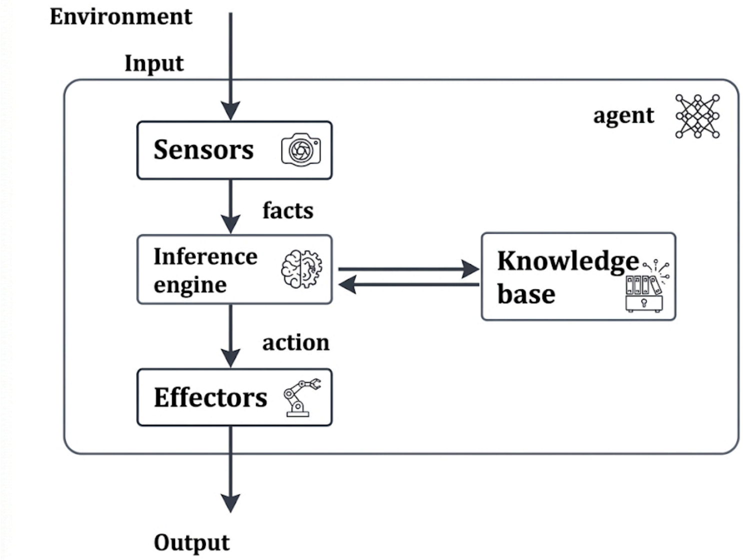
Unit - II Knowledge and Reasoning

2.1 Knowledge-Based Agent in Artificial intelligence

A Knowledge-Based Agent (KBA) is an intelligent agent that uses stored knowledge about the world to make decisions and solve problems.

Example: Medical Diagnosis, KB is Disease or Symptoms.

Architecture



01. Input: Collects environmental data through sensors.

02. Knowledge Base (KB): Stores facts, rules, and domain knowledge.

03. Inference Engine (IE): Applies inference rules to the KB to derive new knowledge and conclusions that support goals.

04. Effectors: Perform actions in the environment based on the conclusions.

05. Example: Medical Diagnosis System. **A) KB:** Contains diseases, symptoms, diagnoses, and treatments. **B) Inference Engine (IE):** Uses rules such as *IF cough AND fever THEN flu*. **C) Effectors/Action:** Displays treatment to the doctor or user.

Terminology

01. Declarative Approach

Knowledge: Represented as facts and declarative statements.

Inference Engine: Draws conclusions by applying logical rules to the facts.

Example: IF Fever(Person) THEN Ill(Person).

Advantage: Easy to represent and modify knowledge.

Disadvantage: Can become inefficient in large systems.

02. Procedural Approach

Knowledge: as procedures or instructions describing how to perform tasks.

Inference Engine: Uses procedures and IF–THEN rules to make decisions.

Example: IF a patient has a fever, THEN suggest a medical test.

Advantage: Efficient for task-oriented problem solving.

Disadvantage: Difficult to explain or trace decision-making logic.

Techniques of Knowledge Representation

01. Logical Representation

- Represents knowledge using facts and logical rules.
- Example: IF cough AND fever THEN flu.
- Advantage: Clear and precise reasoning.
- Disadvantage: Complex for large knowledge bases.

02. Semantic Network Representation

- Represents knowledge as a graph of nodes and relationships.
- Example: Bird → is-a → Animal.
- Advantage: Easy to visualize relationships.
- Disadvantage: Limited reasoning capability.

03. Frame Representation

- Represents knowledge using structured records called frames.
- Example: Person frame → Name, Age, Gender.
- Advantage: Organizes knowledge efficiently.
- Disadvantage: Less flexible for complex reasoning.

04. Production Rule Representation

- Represents knowledge as IF–THEN rules.
- Example: IF fever AND cough THEN flu.
- Advantage: Easy to understand and implement.
- Disadvantage: Large rule sets can be difficult to manage.

Rules of Inference

Logical rules used to derive new conclusions from known facts and statements.

01. Modus Ponens

- If **P → Q** and **P** is true, then **Q** is true.
- **Example:** If fever → illness, fever is present, therefore illness.

02. Modus Tollens

- If **P → Q** and **Q** is false, then **P** is false.
- **Example:** If fever → illness, no illness, therefore no fever.

03. Hypothetical Syllogism

- If **P → Q** and **Q → R**, then **P → R**.
- **Example:** Fever → illness, illness → treatment, therefore fever → treatment.

04. Disjunctive Syllogism

- If **P ∨ Q** and **P** is false, then **Q** is true.
- **Example:** Patient has flu or cold; not flu, therefore cold.

05. Conjunction

- If **P** and **Q** are true, then **P ∧ Q** is true.
- **Example:** Fever and cough → Fever ∧ Cough.

06. Simplification

- If **P ∧ Q** is true, then **P** (or **Q**) is true.
- **Example:** Fever and cough → Fever.

07. Resolution

- From **(P ∨ Q)** and **(¬P ∨ R)**, infer **(Q ∨ R)**.
- **Advantage:** Widely used in automated reasoning and AI systems.

First-Order Logic (FOL)

A knowledge representation technique that extends propositional logic by using **objects, predicates, variables, and quantifiers**.

01. **Predicates:** Describe properties or relationships. Example: Student(John), Likes(Ali, Cricket).
02. **Variables:** Represent objects. Example: x, y.

03. **Quantifiers:** Specify the scope of variables.

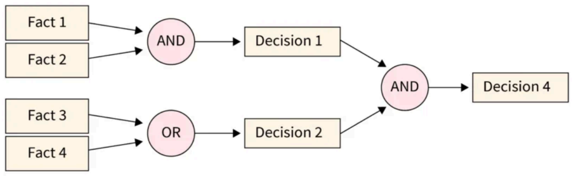
   a. **Universal (∀):** For all objects.

   b. **Existential (∃):** There exists at least one object.

04. **Example:** ∀x (Human(x) → Mortal(x)) → "All humans are mortal."

Forward Chaining

A data-driven inference technique that starts with known facts and applies rules to derive conclusions.



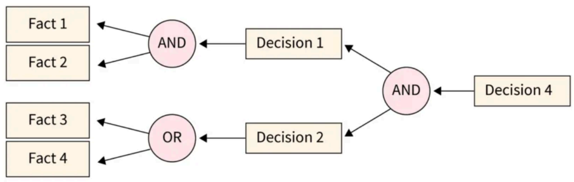
Process: Facts → Rules → New Facts → Conclusion.

Example: **Fact**: Patient has fever and cough. **Rule**: IF fever AND cough THEN flu.

Conclusion: Patient has flu.

Backward Chaining

A goal-driven inference technique that starts with a goal and works backward to find supporting facts.



Process: Goal → Rules → Required Facts → Verification.

Example:

Goal: Determine if the patient has flu.

Rule: IF fever AND cough THEN flu.

Check whether fever & cough are present.

2.2 Reasoning in Artificial intelligence

Definition of Reasoning

Reasoning in AI is the process of using available knowledge and logic to draw conclusions, make decisions, and solve problems.

Types of Reasoning

01. Deductive Reasoning

- Moves from **general rules to specific conclusions**.
- If premises are true, conclusion is guaranteed true.

- **Example:** All humans are mortal → John is human → John is mortal.

02. Inductive Reasoning

- Moves from **specific observations to general rules**.
- Conclusion is probable, not always certain.
- **Example:** Many observed patients with fever have flu → Fever may indicate flu.

03. Abductive Reasoning

- Moves from **observations to the best possible explanation**.
- Used for diagnosis and hypothesis formation.
- **Example:** Patient has cough and fever → Possible flu infection.

04. Analogical Reasoning

- Based on **similarity between two situations**.
- Applies knowledge from a known case to a new case.
- **Example:** A new disease behaves like flu → treat similarly to flu.

2.3 Probabilistic Reasoning in AI

Uncertainty

Uncertainty refers to situations where information is incomplete, vague, or not fully reliable. AI systems cannot always make exact decisions due to missing or noisy data.

Causes of Uncertainty

01. **Incomplete information:** Not all facts are known.
02. **Noisy data:** Sensors or inputs may be inaccurate.
03. **Randomness:** Some events occur unpredictably.
04. **Ambiguity:** Data may have multiple meanings.
05. **Complex environments:** Too many variables to model exactly.

Need of Probabilistic Reasoning in AI

01. Helps AI systems handle **uncertain and incomplete data**.
02. Provides **likelihood-based decisions instead of exact answers**.
03. Improves performance in real-world applications like: Medical diagnosis, Weather prediction, Speech recognition, Autonomous systems
04. Allows reasoning under **risk and uncertainty**.

Bayes' Theorem

A mathematical rule used to update probabilities based on new evidence.

Formula:

Terms:

01. **P(A|B):** Probability of A given B
02. **P(B|A):** Probability of B given A

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

03. **P(A):** Prior probability of A
04. **P(B):** Total probability of B

Example:

A disease affects **1% of people**. A test is **95% accurate**.

- If a person has disease → test is positive with probability **0.95**
- If a person does NOT have disease → test is positive with probability **0.05**

Find the probability that a person has disease given that test is positive.

Step 1: Define

- **A:** Person has the disease
- **B:** Test is positive

Step 2: Given Values

$P(A) = 0.01$

$P(\neg A) = 0.99$

$P(B|A) = 0.95$

$P(B|\neg A) = 0.05$

Step 3: Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Step 4: Find P(B)

$$P(B) = P(B|A)P(A) + P(B|\neg A)P(\neg A)$$

$$P(B) = (0.95 \times 0.01) + (0.05 \times 0.99)$$

$$P(B) = 0.0095 + 0.0495 = 0.059$$

Step 5: Apply Formula

$$P(A|B) = \frac{0.95 \times 0.01}{0.059}$$

$$P(A|B) = \frac{0.0095}{0.059} \approx 0.161$$

Final Answer

$$P(A|B) \approx 0.161\ (16.1\%)$$

Even after a positive test (B), the probability of having disease (A) is only **16.1%** due to low prior probability.